

Oxford Cambridge and RSA

**PREDICTED PAPER**

# **A Level Economics**

**H460/03 Themes in economics**

***Difficulty Level: Advanced***

**Time allowed: 2 hours**

**You can use:**

- a scientific or graphical calculator

## **INSTRUCTIONS**

- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided. If you need extra space use the lined pages at the end of this booklet. The question numbers must be clearly shown.
- Answer all the questions.

## **INFORMATION**

- The total mark for this paper is 80.
- The marks for each question are shown in brackets [ ].
- Quality of extended response will be assessed in questions marked with an asterisk (\*).

## **ADVICE**

- Read each question carefully before you start your answer.

## Section A

Write your answer to each question in the box provided.

**1** A firm owns land, capital equipment, and employs workers. It also pays a royalty to a patent-holder for use of its invention. Which of the following pairings correctly identifies both a factor reward and a transfer payment?

- A Royalty = factor reward to enterprise; patent = transfer payment
- B Rent on land = factor reward; royalty to patent-holder = factor reward to capital
- C Wage to worker = factor reward; unemployment benefit paid by government = transfer payment
- D Wage = transfer payment; rent = factor reward to enterprise

Your answer  [1]

**2** In a closed economy with no government sector, the marginal propensity to consume is 0.75. The government now introduces a proportional income tax with an effective marginal rate of 20%. Autonomous investment rises by £100bn. What is the final change in national income?

- A £160 billion
- B £250 billion
- C £400 billion
- D £500 billion

Your answer  [1]

**3** An economy is in a liquidity trap. Nominal interest rates are close to zero and cannot fall further. Which of the following policy tools is LEAST likely to stimulate aggregate demand?

- A A large-scale fiscal stimulus financed by government borrowing
- B A reduction in the central bank's base rate by 0.25 percentage points
- C Forward guidance committing the central bank to keep rates low for longer
- D Quantitative easing aimed at reducing long-term bond yields

Your answer  [1]

**4** A country's central bank sells government bonds on the open market. Through the monetary transmission mechanism, the most likely short-run sequence of effects is:

- A Bond prices rise → yields fall → borrowing rises → AD rises
- B Bond prices fall → yields rise → bank reserves fall → lending falls → AD falls
- C Money supply rises → inflation rises → exchange rate falls → exports rise
- D Money supply rises → interest rates fall → investment rises → AD rises

Your answer  [1]

**5** Goods X and Y are complements in production (joint supply). The XED for X with respect to Y (consumption) is  $-0.4$ . An advertising campaign for good Y increases demand for Y by 20%. Assuming the supply curves of both goods are upward-sloping and PED<sub>S</sub> is the same for both, what is the likely direction of impact on the price of good X?

- A Price of X falls because joint supply increases the supply of X
- B Price of X rises because X and Y are complements in demand
- C Price of X rises by exactly 8%
- D Price of X is unaffected

Your answer  [1]

**6** Two firms dominate a homogeneous oligopoly market and engage in Cournot (quantity) competition. The market inverse demand function is  $P = 100 - Q$ , and both firms have constant marginal cost of £10. The Cournot-Nash equilibrium output for each firm is:

- A 15 units each; market price = £70
- B 22.5 units each; market price = £55
- C 30 units each; market price = £40
- D 45 units each; market price = £10

Your answer ☐ [1]

**7** Digital platforms exhibit strong network effects: a user's utility rises with the total number of users. This feature makes the long-run market structure tend towards:

- A Monopolistic competition with heavy product differentiation
- B Perfect competition as entry barriers fall over time
- C Tipping towards monopoly or dominant-firm outcomes ('winner-takes-most')
- D A stable oligopoly with equal market shares

Your answer ☐ [1]

**8** A monopolist faces inverse demand  $P = 120 - 2Q$  and total cost  $TC = 20 + 20Q$ . What is the monopolist's profit-maximising output, price, and supernormal profit?

- A  $Q = 20$ ,  $P = £80$ , profit = £1,180
- B  $Q = 25$ ,  $P = £70$ , profit = £1,230
- C  $Q = 30$ ,  $P = £60$ , profit = £1,180
- D  $Q = 50$ ,  $P = £20$ , profit = £0

Your answer ☐ [1]

**9** A consumer's total utility from apples follows the function  $TU = 10Q - 0.5Q^2$ . At what quantity does diminishing marginal utility first occur, and at what quantity is total utility maximised?

- A Diminishing MU from  $Q=1$ ; TU max at  $Q=10$
- B Diminishing MU from  $Q=2$ ; TU max at  $Q=10$
- C Diminishing MU from  $Q=1$ ; TU max at  $Q=20$
- D Diminishing MU never occurs; TU max at infinity

Your answer ☐ [1]

**10** A UK firm buys \$1,000,000 of machinery from the US. The spot exchange rate is £1 = \$1.25. The firm simultaneously enters a forward contract to sell dollars and buy pounds in 90 days at £1 = \$1.20. If the spot rate in 90 days is £1 = \$1.30, what is the net cost in pounds for the machinery, and did the forward contract save or cost the firm money (compared with spot)?

- A Cost £800,000; forward saved £30,000 vs spot
- B Cost £833,333; forward cost £64,103 vs spot
- C Cost £800,000; forward cost £30,000 vs spot
- D Cost £769,231; forward saved £64,103 vs spot

Your answer ☐ [1]

**11** A profit-maximising monopolist with constant marginal cost practises third-degree price discrimination across two separate markets. Market A has  $PED = -2$  at the relevant price; Market B has  $PED = -4$ . Using the inverse-elasticity pricing rule, the ratio of the mark-up over MC in Market A compared with Market B is:

- A 1 : 1 (equal mark-ups)
- B 2 : 1 (A has twice the mark-up of B)
- C 3 : 2 (Market A has 1.5× the mark-up of B)
- D 1 : 2 (B has twice the mark-up of A)

Your answer ☐ [1]

**12** Country X and Country Y have the same labour force of 10 million. In Country X, output per worker is £80,000 and annual productivity growth is 2%. In Country Y, output per worker is £50,000 and annual productivity growth is 4%. Assuming constant labour force and all else equal, after how many years will Country Y's output per worker exceed Country X's? (to nearest year)

- A 20 years
- B 24 years
- C 32 years
- D Never, because X starts higher

Your answer ☐ [1]

**13** A monopsony employer faces a labour supply curve  $W = 10 + 0.5L$  (where  $W$  is the wage and  $L$  is the quantity of labour) and a marginal revenue product curve  $MRP = 70 - L$ . The marginal cost of labour is  $MCL = 10 + L$ . What is the profit-maximising wage and level of employment?

- A  $W = £10, L = 0$
- B  $W = £20, L = 20$
- C  $W = £25, L = 30$
- D  $W = £40, L = 60$

Your answer ☐ [1]

**14** An economy is described by the Fisher equation  $MV = PY$ . Money supply grows at 8% per year, the velocity of circulation is stable, and real GDP grows at 3% per year. According to monetarist theory, what is the long-run rate of inflation?

- A 0%
- B 3%
- C 5%
- D 11%

Your answer ☐ [1]

**15** A country with a floating exchange rate experiences a sudden 30% depreciation.  $PED_X = -0.3$  and  $PED_M = -0.4$ . According to the Marshall-Lerner condition, the short-run effect on the current account balance will be:

- A Improvement because the sum of elasticities is positive
- B Improvement of approximately 21% × volumes
- C No change because depreciations are neutral
- D Worsening because  $|PED_X| + |PED_M| < 1$

Your answer ☐ [1]

**16** Country Z has a flat income tax of 25% on all income above a personal allowance of £15,000. Taxpayer P earns £20,000; taxpayer Q earns £80,000. Their respective average tax rates are:

- A P: 6.25%; Q: 20.3% — the tax is progressive

- B P: 25%; Q: 25% — the tax is proportional
- C P: 6.25%; Q: 20.3% — the tax is regressive
- D P: 1.25%; Q: 25% — the tax is proportional

Your answer ☐ [1]

**17** The Lorenz curve of Country A lies closer to the line of perfect equality than the Lorenz curve of Country B. However, Country A's Gini coefficient is HIGHER than Country B's. This is:

- A Consistent — closer to equality means higher Gini
- B Impossible — a Lorenz curve closer to the line of equality must produce a lower Gini
- C Possible if Country B's population is much larger
- D Possible if the Lorenz curves cross each other

Your answer ☐ [1]

**18** A country introduces a specific (per-unit) carbon tax. If the demand for the taxed fuel is highly inelastic and supply is highly elastic, the incidence of the tax falls:

- A Entirely on the government
- B Largely on consumers; government revenue is large
- C Largely on producers; government revenue is small
- D Equally on consumers and producers regardless of elasticities

Your answer ☐ [1]

**19** An open economy has the following annual flows (£ billions): exports = 400; imports = 450; investment = 180; savings = 150; government spending = 220; taxation = 200. Using the circular-flow identity, the economy has:

- A A net injection of £20bn — national income will rise
- B A net leakage of £20bn — national income will fall
- C Equilibrium — injections equal leakages
- D Cannot be determined without the MPC

Your answer ☐ [1]

**20** Asymmetric information leads to adverse selection in the second-hand car market. Which of the following best describes the welfare effect described by Akerlof's 'lemons' model?

- A The average price of used cars rises, pushing out buyers
- B High-quality cars are progressively withdrawn from the market; only low-quality 'lemons' remain, reducing welfare
- C Prices rise above the socially optimal level due to hidden action after the sale
- D The market reaches an efficient outcome via signalling alone

Your answer ☐ [1]

**21** A perfectly competitive industry is in long-run equilibrium. The government imposes a lump-sum tax on every firm. In the short run and the long run respectively, the industry output and the number of firms will:

- A Short run: unchanged; long run: some firms exit, industry output falls
- B Short run: output falls; long run: output returns to original level
- C Short run: output rises; long run: firms enter
- D Unchanged in both short and long run

Your answer ☐ [1]

**22** A dominant digital platform engages in 'self-preferencing' — prioritising its own goods in search results over those of third-party sellers on the same platform. This practice is MOST similar to which classical abuse of market power?

- A First-degree price discrimination
- B Predatory pricing to eliminate competitors
- C Vertical foreclosure / leveraging upstream dominance into a downstream market
- D Tacit collusion between oligopolists

Your answer ☐ [1]

**23** A small open economy imposes a tariff on imports. Which of the following effects is NOT necessarily present?

- A A deadweight welfare loss from production distortion
- B A deadweight welfare loss from consumption distortion
- C A rise in government tariff revenue
- D An improvement in the country's terms of trade

Your answer ☐ [1]

**24** The money multiplier in a fractional-reserve banking system is approximately 10. The central bank undertakes £50bn of quantitative easing, crediting commercial bank reserves. If banks hold 30% of the new reserves as excess (idle) reserves and lend out the rest, the final expansion of broad money will be approximately:

- A £50 billion
- B £150 billion
- C £350 billion
- D £500 billion

Your answer ☐ [1]

**25** A government imposes a binding minimum price (price floor) on a good where demand is inelastic and supply is elastic. Which of the following is NOT a consequence?

- A Emergence of an excess supply (surplus)
- B Producer surplus unambiguously rises
- C A deadweight welfare loss occurs
- D Consumer surplus falls

Your answer ☐ [1]

**26** A perfectly contestable market has zero sunk costs but moderate economies of scale. In long-run equilibrium, the incumbent firm(s) must produce at:

- A Minimum average cost, earning normal profit
- B The allocatively efficient point ( $P = MC$ ) with supernormal profit
- C Below minimum efficient scale, earning a loss
- D The output where  $MR = MC$ , earning supernormal profit sustained by sunk costs

Your answer ☐ [1]

**27** A negative supply-side shock (e.g., rising global oil prices) and an expansionary fiscal stimulus occur simultaneously. In an AD/AS framework with an upward-sloping SRAS, the short-run effect on real GDP and the price level is:

- A Real GDP unambiguously rises; price level rises unambiguously
- B Real GDP change is ambiguous; price level rises unambiguously
- C Real GDP falls unambiguously; price level change is ambiguous
- D Both real GDP and the price level fall

Your answer ☐ [1]

**28** Country W has the following HDI components: life expectancy index = 0.90; education index = 0.85; income (GNI per capita) index = 0.95. Country V has 0.95, 0.95, and 0.75 respectively. Using the geometric mean formula  $HDI = (I_{life} \times I_{edu} \times I_{income})^{(1/3)}$ , which country has the higher HDI (to 3 d.p.)?

- A Country W, with HDI = 0.899
- B Country V, with HDI = 0.876
- C Country W, with HDI = 0.900; Country V, HDI = 0.876
- D They are equal at HDI  $\approx$  0.888

Your answer ☐ [1]

**29** A developing country with a large informal/agricultural sector is industrialising. According to the Lewis two-sector model, in the early stages of industrialisation:

- A Urban wages rise rapidly due to scarce labour
- B Urban wages remain stable at a small premium above subsistence; profits in the modern sector rise and are reinvested
- C Rural wages rise sharply due to out-migration
- D HDI falls as inequality rises

Your answer ☐ [1]

**30** Which of the following transactions appears as a CREDIT on the UK primary income account?

- A A UK tourist buying a meal in Spain
- B A UK-resident shareholder receiving dividends from a French company
- C A UK charity sending foreign aid to Somalia
- D A UK importer paying for German machinery

Your answer ☐ [1]

## **Section B**

*Read the extracts before answering the questions.*



**Extract 1*****Big Tech, network effects and digital-market regulation***

In the first quarter of 2025, the world's five largest publicly-listed companies by market capitalisation — Apple, Microsoft, Alphabet (Google), Amazon, and Meta (Facebook/Instagram) — were collectively valued at over US\$12 trillion, more than the GDP of any country except the United States and China. The competition regulators of the United States, the European Union, and the United Kingdom have in parallel launched a wave of investigations and enforcement actions targeting the market power of these 'Big Tech' firms.

Digital markets are characterised by strong network effects (the utility to a user rises as more users join a platform), extensive economies of scale (software development costs are largely fixed), and economies of scope (data gathered from one service can be used to improve another). These properties tend to produce 'winner-takes-most' markets in which a single firm — or at most two — captures the lion's share of demand. By 2024 Google held 91% of the UK search market; Apple's App Store and Google Play between them handled 99% of all UK smartphone app downloads; and Meta's services (Facebook, Instagram, WhatsApp) reached over 80% of UK internet users weekly.

The UK's Digital Markets, Competition and Consumers Act 2024 grants the Competition and Markets Authority (CMA) new powers to designate firms with 'Strategic Market Status' (SMS) and impose tailored conduct requirements — for example requiring Apple to open up its App Store to third-party payment systems, or requiring Google to give equal search-ranking treatment to competing mapping or shopping services. The EU's Digital Markets Act (DMA) has similar aims and designates firms as 'gatekeepers'. Fines under the UK regime can reach 10% of global turnover.

Big Tech firms argue that their dominance reflects genuine consumer benefits, not anti-competitive conduct: their services are mostly free at the point of use, deliver dynamic efficiencies through continual innovation (the iPhone in 2007, cloud computing, generative AI in 2023–25), and their profit is reinvested into research and development. They also warn that over-regulation will harm consumers by fragmenting ecosystems, creating security vulnerabilities, and reducing innovation incentives. Critics respond that free-at-point-of-use does not mean costless: users 'pay' with their personal data, and small rivals and app developers are squeezed by platform fees (typically 15–30% on the App Store) and self-preferencing in search rankings.

**Fig. 1.1: Market capitalisation of the five largest Big Tech firms, 2019 vs 2024 (US\$ trillion)**

**Apple: 1.30 → 3.50   Microsoft: 1.20 → 3.10   Alphabet: 0.90 → 2.05**  
**Amazon: 0.95 → 1.85   Meta: 0.58 → 1.35   Combined: 4.93 → 11.85**

**Fig. 1.2: Research & Development spending as % of revenue (2023)**

**Alphabet: 14.1%   Meta: 27.0%   Microsoft: 12.8%   Apple: 7.8%**  
**Typical S&P 500 firm: 4.2%   UK FTSE 100 average: 2.1%**

**31** Using the data in Fig. 1.1, calculate the percentage increase in the combined market capitalisation of the five Big Tech firms between 2019 and 2024.

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**32** Explain, using examples from Extract 1, two reasons why digital markets tend towards 'winner-takes-most' outcomes.

**[3]**

**33\*** Evaluate, using an appropriate diagram and the information in Extract 1, whether increased regulation of Big Tech firms is likely to improve economic welfare. [15]

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Extract 2

Growth, development and inequality in emerging economies

In 2024 India overtook Japan to become the world's fourth-largest economy in nominal US dollar terms, with GDP of approximately \$4.1 trillion, behind only the United States, China and Germany. Over the past three decades India's real GDP has grown at an average of around 6.2% per year, while Vietnam — a country of under 100 million people — has grown at around 6.5% per year, lifting its GNI per capita (PPP) from around \$1,200 in 1993 to approximately \$14,500 in 2024.

Growth of this scale has transformed living standards. India's Human Development Index rose from 0.434 in 1990 to 0.644 in 2022, and Vietnam's from 0.482 to 0.726 over the same period — Vietnam now classed as 'high' human development. Life expectancy in India is 72 years; in Vietnam 74 years. But growth has been uneven. India's Gini coefficient for wealth has risen from 0.54 in 2000 to 0.65 in 2022 — one of the highest in the world. The top 1% of earners in India captured around 22% of national income in 2022. Vietnam's Gini is lower at 0.36 but also rising.

Both economies have moved large parts of their labour forces out of low-productivity agriculture. India's agricultural share of employment fell from 60% in 1993 to 44% in 2022; Vietnam's from 70% to 27%. In line with the Lewis two-sector model, surplus rural labour has been absorbed by urban manufacturing and service sectors, driving productivity growth — though rural-urban wage gaps remain large (Vietnam's average urban wage is roughly 1.8 times the rural wage in 2023).

Challenges persist. Around 230 million people in India live below the national poverty line. Access to clean water, sanitation and quality education is uneven. Vietnam remains heavily dependent on foreign direct investment — FDI inflows reached \$36bn in 2023 (around 8% of GDP). Both economies face the 'middle-income trap': the risk that rapid catch-up growth slows as cheap-labour advantages erode but domestic innovation and human capital remain insufficient to compete with advanced economies. Their governments have responded with large public investment in infrastructure, education and R&D, and in India's case, targeted 'Production Linked Incentive' schemes to grow domestic manufacturing.

Fig. 2.1: Selected economic indicators, 2024

|                           | India | Vietnam | UK (for comparison) |
|---------------------------|-------|---------|---------------------|
| GDP per capita (PPP, \$): | 9,200 | 14,500  | 54,000              |
| Real GDP growth (%):      | 6.5   | 6.0     | 0.8                 |
| HDI:                      | 0.644 | 0.726   | 0.940               |
| Gini coefficient:         | 0.65  | 0.36    | 0.35                |
| Agriculture % workforce:  | 44    | 27      | 1                   |

Fig. 2.2: India — HDI and GDP per capita (PPP, \$), 1990 vs 2022

1990: HDI 0.434, GDP per capita \$1,180    2022: HDI 0.644, GDP per capita \$8,790

34 Using the data in Fig. 2.1 and Fig. 2.2, calculate (a) the percentage change in India's HDI and (b) the percentage change in India's GDP per capita (PPP) between 1990 and 2022.

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**[3]**

**35** Using Fig. 2.1, identify TWO limitations of using HDI alone to compare living standards across the three countries.

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[2]

**36\*** Evaluate, using the information in Extract 2, whether rapid economic growth is the most effective route to raising living standards in emerging economies such as India and Vietnam. [15]

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Extract 3

Gig-economy workers, platform monopsony and employment status

By 2024 an estimated 4.7 million people in the UK (approximately 14% of the workforce) had earned income from digital gig-economy platforms — including Uber, Deliveroo, Amazon Flex, Bolt, and others. Globally the total is over 435 million. Gig workers are typically classified by platforms as 'independent contractors' rather than 'employees' or the intermediate UK legal category of 'workers'. This classification determines whether workers are entitled to the National Living Wage, paid holiday, sick pay, employer pension contributions, and protection against unfair dismissal.

In the landmark case *Uber BV v Aslam* (UK Supreme Court, 2021), the court ruled that Uber drivers are 'workers' — entitling them to the NLW, holiday pay, and rest breaks while logged into the app. The ruling has been extended to other platforms through subsequent litigation but has not stopped many platforms from attempting to reclassify roles. Analysts estimate that the 2021 ruling increased Uber's UK operating costs by approximately 15%, some of which has been passed through to consumers in higher fares.

Gig platforms are often described as exercising 'algorithmic monopsony' over drivers and couriers. Although a driver can in principle work for several apps, in practice most drivers derive over 70% of their earnings from a single platform, and switching is deterred by ratings systems, acceptance-rate requirements, and area-specific demand. Platforms set pay rates unilaterally through opaque algorithms; drivers cannot collectively bargain in the traditional sense because they are classed as self-employed for tax purposes.

Supporters argue that gig work provides flexibility, low barriers to entry, and access to income for students, carers, and those excluded from traditional employment. A 2023 survey found that 62% of UK gig workers cited 'flexibility' as the main reason for taking the work. Critics argue the flexibility is illusory because algorithms allocate work, income is unpredictable, and effective hourly earnings (net of fuel, vehicle costs, and unpaid waiting time) can fall below the NLW. A 2024 University of Oxford study found median net pay for UK Uber drivers, after deducting vehicle costs, was £6.20 per hour — below the NLW of £11.44.

Fig. 3.1: UK gig-economy workforce and average hourly earnings (after costs)

| Year | Workers (m) | Median net pay/hr | NLW    |
|------|-------------|-------------------|--------|
| 2016 | 1.1         | £7.40             | £7.20  |
| 2019 | 2.8         | £7.60             | £8.21  |
| 2022 | 4.4         | £6.40             | £9.50  |
| 2024 | 4.7         | £6.20             | £11.44 |

37 Using Fig. 3.1, compare the change in median net hourly pay for gig workers with the change in the NLW between 2016 and 2024.

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[2]

**38** Evaluate, using the information in Extract 3, whether gig-economy workers should be reclassified as full employees entitled to the National Living Wage, paid holiday and employer pension contributions. **[8]**

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**END OF QUESTION PAPER**